Math 421

Monday, November 24

Announcements

- This week's drop-in hours:
  - **Tuesday** 9-11 am VV B205
  - Wednesday 12-1 pm VV 311
- Check for any final exam conflicts & email me

Differentiable Implies Continuous

Theorem 28.2. If f is differentiable at a, then f is continuous at a.

Discussion. Given: $\lim_{x \to a} \frac{f(x) - f(a)}{x - a} = f'(a)$ Want to show: $\lim_{x \to a} f(x) = f(a)$

$$\lim_{x \to a} f(x) = \lim_{x \to a} \left[(x-a) \frac{f(x) - f(a)}{x - a} + f(a) \right]$$

$$= \lim_{x \to a} (x-a) \cdot \lim_{x \to a} \frac{f(x) - f(a)}{x - a} + \lim_{x \to a} f(a)$$

$$= 0 \cdot f'(a) + f(a)$$

$$= f(a).$$

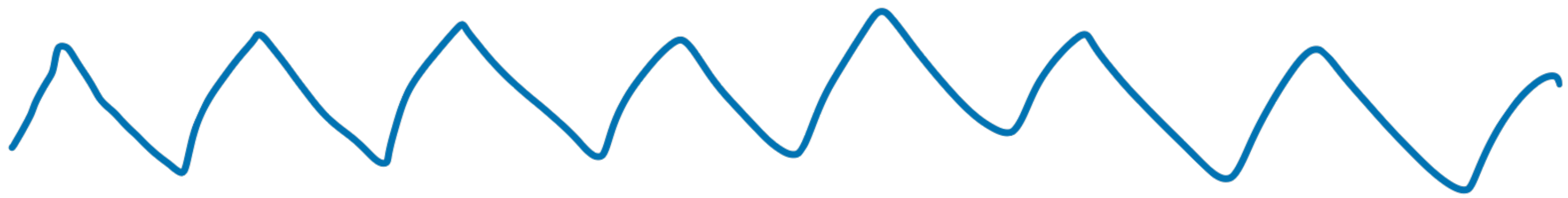
Discussion Questions

Q1: Is the converse of the theorem true? No.

$f(x) = |x|$

Q2: There exist functions that are *continuous everywhere* but *differentiable nowhere*. These are difficult to construct, but can you try to describe what it might look like? It could be helpful to first think of the following things:

- What might it look like for a function to be continuous but not differentiable at a *finite number of points*?
- What might it look like for a function to be continuous but not differentiable at *every integer*?



Algebraic Derivative Rules

Theorem 28.3. Let f and g be differentiable at $x = a$ and let $c \in \mathbb{R}$. Then

i. $(cf)'(a) = cf'(a)$.

ii. $(f + g)'(a) = f'(a) + g'(a)$.

$(cf)(x) = c \cdot f(x)$

$(f+g)(x) = f(x) + g(x)$

iii. [product rule] $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$.

iv. [quotient rule] if $g(a) \neq 0$, then $\left(\frac{f}{g}\right)'(a) = \frac{g(a)f'(a) - f(a)g'(a)}{\left(g(a)\right)^2}$.

Given: $\lim_{x\to a} \frac{f(x)-f(a)}{x-a} = f'(a)$ $\quad \lim_{x\to a} \frac{g(x)-g(a)}{x-a} = g'(a)$

Activity

Prove parts i & ii using the limit definition of the derivative and the algebraic limit theorems.

i. $(cf)'(a) = cf'(a)$.

$$\lim_{x\to a} \frac{(cf)(x) - (cf)(a)}{x-a} = \lim_{x\to a} \frac{cf(x) - cf(a)}{x-a} = c \cdot \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$$
$$= cf'(a)$$

ii. $(f+g)'(a) = f'(a) + g'(a)$.

$$\lim_{x\to a} \frac{(f+g)(x) - (f+g)(a)}{x-a} = \lim_{x\to a} \frac{f(x)+g(x) - (f(a)+g(a))}{x-a}$$
$$= \lim_{x\to a} \left[\frac{f(x)-f(a)}{x-a} + \frac{g(x)-g(a)}{x-a} \right] = f'(a) + g'(a)$$

Given: $\lim_{x \to a} \frac{f(x) - f(a)}{x - a} = f'(a)$ $\lim_{x \to a} \frac{g(x) - g(a)}{x - a} = g'(a)$

Proof of Product Rule

Theorem 28.3. (iii) Let f and g be differentiable at $x = a$ and let $c \in \mathbb{R}$. Then $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$.

Discussion.

$$\lim_{x \to a} \frac{(fg)(x) - (fg)(a)}{x - a} = \lim_{x \to a} \frac{f(x)g(x) - f(a)g(a)}{x - a}$$

Q: what expression can we add & subtract to the numerator to introduce the quotients we were given?

$$= \lim_{x \to a} \frac{f(x)g(x) - g(x)f(a) + g(x)f(a) - f(a)g(a)}{x - a}$$

$$= \lim_{x \to a} \left[g(x) \cdot \frac{f(x) - f(a)}{x - a} + f(a) \cdot \frac{g(x) - g(a)}{x - a} \right] = \boxed{g(a)} f'(a) + f(a)g'(a).$$

differentiable functions are continuous

Ross: uses limit definition & binomial theorem

Power Rule

Theorem. For all natural numbers $n \in \mathbb{N}$, if $f(x) = x^n$, then $f'(x) = nx^{n-1}$.

Activity: Prove the power rule.

Q: What proof technique could you use? Induction. & use product rule.

Base Case: If $n=1$ $f(x)=x$ $f'(x)=1=1\cdot x^{1-1}=1$ ✓.

Induction Step: Assume for some n, if $f(x)=x^n$, then $f'(x)=nx^{n-1}$.

Then for $f(x)=x^{n+1}=x\cdot x^n$, so

$f'(x)=x\cdot(nx^{n-1})+x^n\cdot 1=nx^n+x^n=(n+1)x^n$

Activity – Negative Power Rule

Let m be a positive integer, and let $h(x) = x^{-m}$ for $x \neq 0$. Then $h(x) = f(x)/g(x)$ where $f(x) = 1$ and $g(x) = x^m$. Use the quotient rule to find $h'(a)$.

$h(x) = \frac{1}{x^m}$ $\qquad$ $h'(a) = \frac{x^m \cdot 0 - 1 \cdot m x^{m-1}}{(x^m)^2}$

$= \frac{-m x^{m-1}}{x^{2m}} = -m x^{-m-1}$